MINUTE READ

Deconvolution of capacitive and faradaic current

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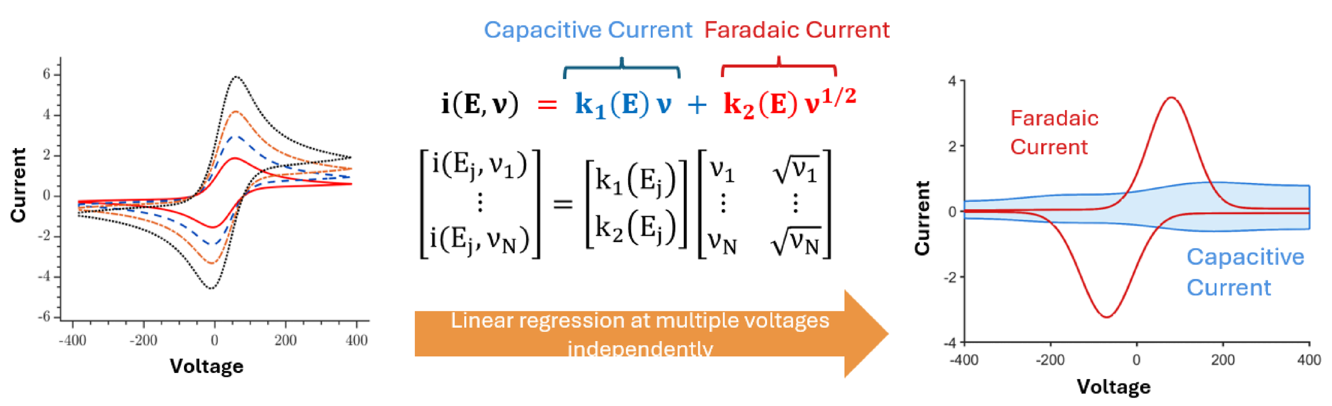
THE PROBLEM

The $\nu/\nu^{1/2}$ method splits current into a capacitor-like part that scales with scan rate (ν) (non-faradaic current) and a battery-like part (faradaic current) that scales with its square root ($\nu^{1/2}$). It's widely used to classify energy-storage materials; but rarely tested against a known answer.



WHAT WE DID

We chose the simplest, best-understood case. One clean redox reaction + planar electrode and checked the split two ways: in simulation (where the exact answer is known) and in experiment (ferrocene as active species, capacitive current measured directly from the electrolyte)

WHAT WE FOUND

It mostly fails. The split is valid only under highly restricted conditions; elsewhere it reports unphysical and spurious currents while a near-perfect fit ($R^2 \approx 1$) hides it.

ROOT CAUSE

The true scan-rate scaling isn't fixed at $\nu^{1/2}$; a local Damköhler number (Da) sets it point by point, and $\nu^{1/2}$ holds only in the fast-reaction (electrochemically reversible) limit.

POSSIBLE SOLUTION

Fit physics based forward model across all scan rates at once: recovering physically consistent deconvolution.

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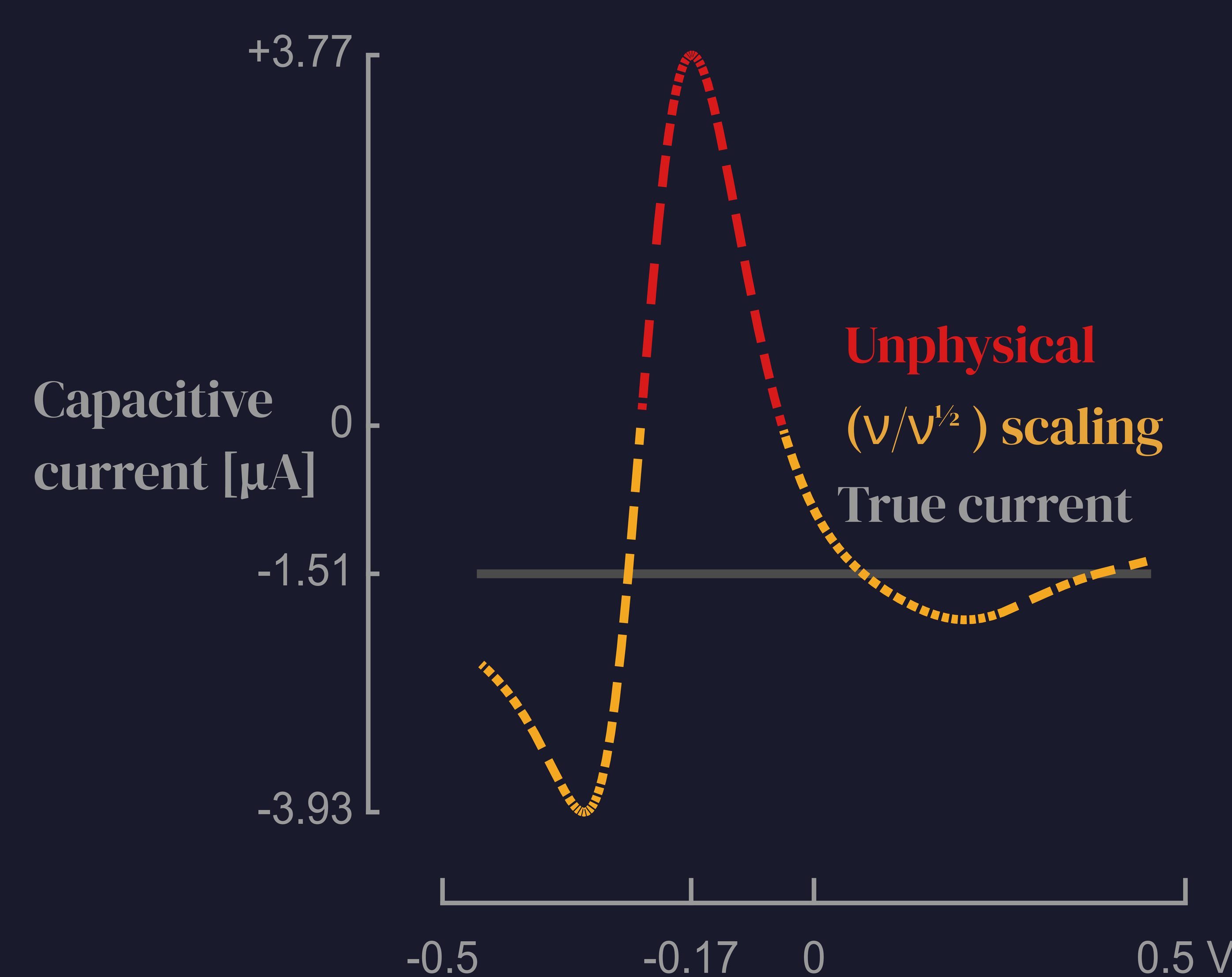
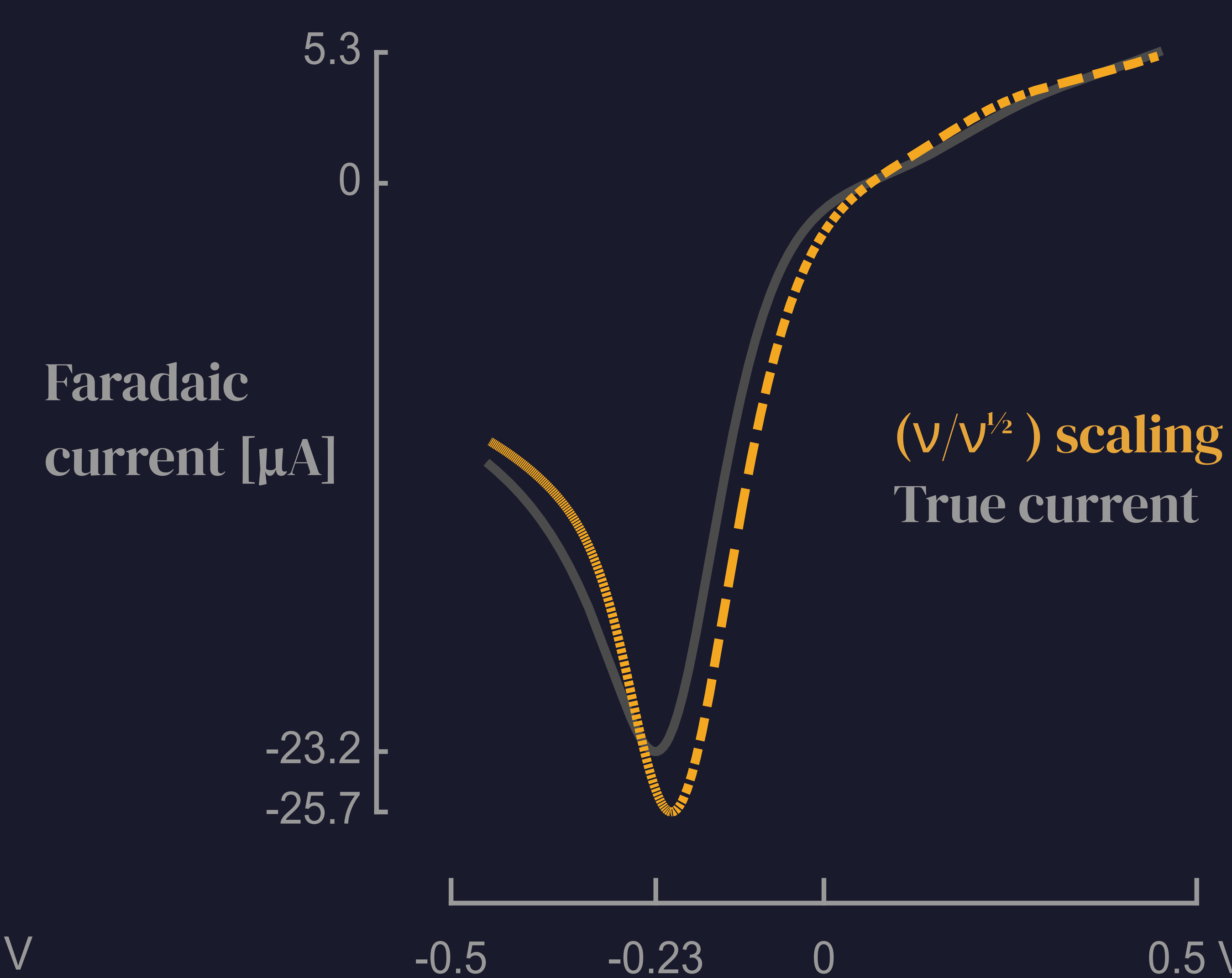
Standard way to deconvolute charge storage currents is asymptotic $Da \to \infty$: it fails at finite Da and perfect fit ($R^2 \approx 1$) misleads.

Capacitive current fitted by $\nu/\nu^{1/2}$ scaling gives huge errors and unphysical values; a flat double layer response is recast as large peak of the wrong sign

Faradaic current's fit is systematically distorted and overshot; as fixed $\nu^{1/2}$ basis cannot capture a scan rate exponent that varies with potential





Tap (left-NFC) or scan (right-QR) to see inside the baseball -- interactive plots and derivations

Scan to know more about the layout